SPINAL RED FLAGS CASE STUDIES

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Introduction

International Framework for Red Flags for Potential Serious Spinal Pathologies

The framework has been developed by researchers and clinicians to provide a pragmatic approach for clinicians to screen for serious spinal pathology that can masquerade as musculoskeletal spinal conditions.

The framework has been informed by available evidence and augmented by a formal consensus process that included academics and clinicians involved in the management of musculoskeletal conditions.

How to use the framework

Clear and open communication with people with potential serious pathology is vital.

Provide reassurance about why you are assessing for red flags, especially when the person is likely at low risk of having severe pathology.

Consider the wording of your questions, your body language, tone of voice, and mannerisms when asking the questions.

People must feel at ease when answering questions and not judged (eg, intravenous drug use).

Ensure people have sufficient time to consider and communicate their answers about something that they may never have considered before.

Provide support regarding the emotional impact of being assessed for potentially life-changing conditions.

Help people find other services that can offer further support.

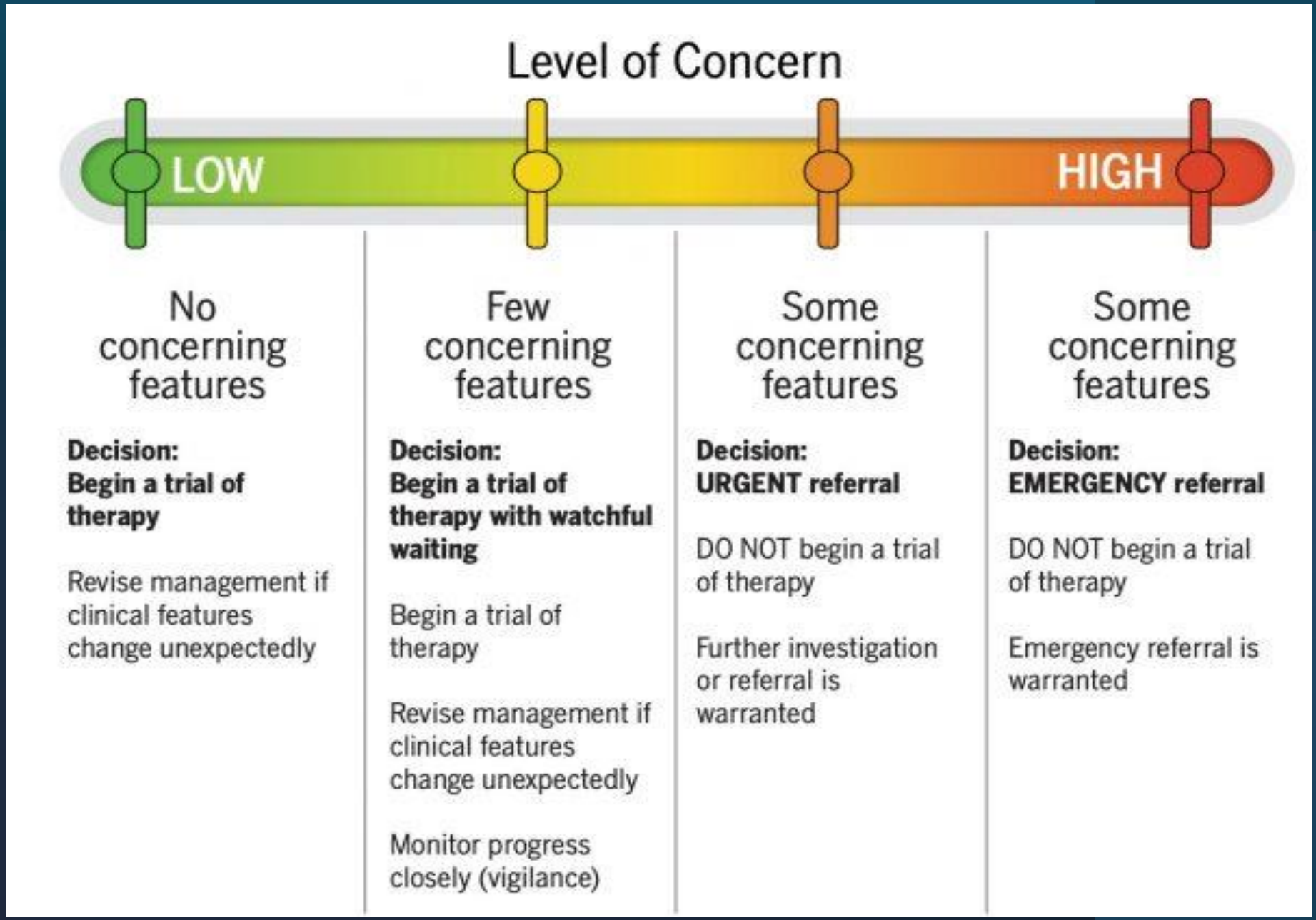
Red Flags

Red flags are signs and symptoms that raise suspicion of serious spinal pathology.

For spinal pathology, 163 signs and symptoms have been reported as red flags, including 119 symptoms from the individual's history and 44 signs from the physical examination.

How to decide??

1. Determine your level of concern. Consider the evidence to support red flags and the individual profile of the person's health determinants (eg, age, sex) to decide your level of concern (your index of suspicion) about the presence of serious pathology.
2. Decide on your clinical action, based on your level of concern determined in step 1.
3. Consider the pathway for emergency/urgent referral.
4. Know your local referral pathways and pathways to access specialist care if indicated


Level of Concern
LOW
HIGH
No concerning features
Decision: Begin a trial of therapy
Revise management if clinical features change unexpectedly
Few concerning features
Decision: Begin a trial of therapy with watchful waiting
Begin a trial of therapy
Revise management if clinical features change unexpectedly
Monitor progress closely (vigilance)
Some concerning features
Decision: URGENT referral
DO NOT begin a trial of therapy
Further investigation or referral is warranted
Some concerning features
Decision: EMERGENCY referral
DO NOT begin a trial of therapy
Emergency referral is warranted

Cauda Equina

If you suspect CES, perform a full neurological assessment to establish dermatome sensory loss, myotome weakness, or reflex changes.

A digital rectal examination is no longer considered essential in a primary care setting in the United Kingdom. It is necessary in secondary care to evaluate loss of anal sphincter tone.

Sensation to light touch and pinprick throughout the saddle region, including the buttocks, inner thighs, and perianal region, is a necessary test in any clinical situation (chaperone).

People should be sent for emergency magnetic resonance imaging (MRI) and surgical opinion. Positive findings are likely to be accompanied by an ultrasound of the bladder.

Case 1

- A woman is urgently referred to you with back pain. Within the wider detailed subjective and objective examination, there are no positive items on the CES cue card.
- LBP, no leg pain
- No symptoms of CES
- Level of Concern?

- The woman has back and leg pain. The pain in her leg is getting worse and is now radiating distally below the knee, and she has started to notice pain in the other leg.
- Nothing on CES questioning is positive, and there is no existing neurological deficit.
- Leg pain worsening
- Signs of bilateral leg pain
- Level of Concern?
- Clinical action?

Contin. Case 1

- The woman now has back and increasing leg pain bilaterally. She reports one episode of incontinence 4 weeks ago, but not since. Neurological examination is unremarkable.

- Back and bilateral leg pain increasing
- One episode of incontinence 4 weeks ago
- Level of concern?
- Clinical action?

The woman has now developed a 1-week history of some numbness of the left side of the vagina.

Back and bilateral leg pain increasing

One episode of incontinence 4 weeks ago

One-week history of vaginal numbness

- Level of concern?
- Clinical action?

Spinal Fractures

- While these are predominantly a risk for older patients, especially women, clinicians need to be aware of the risk factors and signs and symptoms of spinal fractures and to consider what detailed questions to ask to help with further management.



Case 2

- A 35-year-old man presents with sudden onset of thoracic pain after lifting a heavy bag of concrete. The man has no previous history of fracture and is generally in good health. He smokes 5 cigarettes a day and has done so for 10 years. He has limited thoracic spine movement into rotation to both sides. He is locally tender to palpation at T8 and T9 unilaterally on both sides. He denied family history of spinal fractures, steroids use, previous fractures.
- **Level of Concern?**
- **Clinical action?**

Case 3

- A 78-year-old woman presents with upper lumbar pain. No precipitating injury was reported, but the pain has worsened over the last 3 months. The pain is worse when lying supine. She has a history of left radius fractures. She had her menopause at age 38, having started her periods at 15 years of age. She is otherwise well and has no family history of osteoporosis.
- **Level of concern?**
- **Clinical action?**

Case 4

- A 60-year-old woman presents with moderately severe thoracolumbar pain after bending over and lifting a heavy pot in the garden 3 weeks ago. Her pain is slightly improved. She is otherwise well and not on any medication other than paracetamol for her pain.
- She has no history of fracture. She had an early menopause at age 35. She smokes 20 cigarettes per day. She has pain in extension and rotation, some local spinal tenderness, and zygapophyseal (facet) joint tenderness bilaterally.
- **Level of Concern?**
- **Clinical action?**

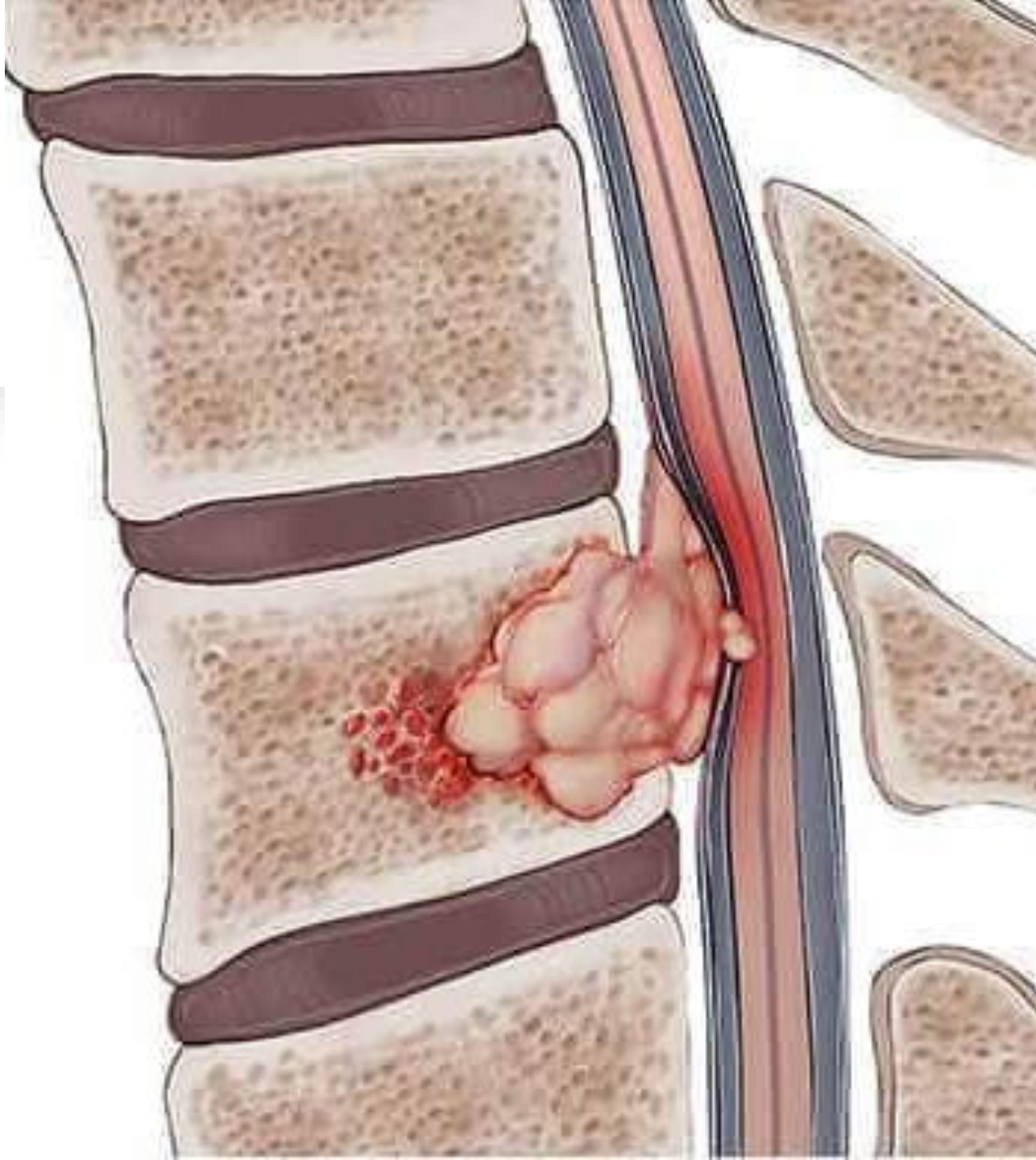
Case 5

- A 74-year-old man with mid-thoracic and lumbar pain presents with increasing pain locally in the spine, but no trauma/injury. Pain is worse in lying and standing and eased slightly in sitting. He is taking increasing doses and strengths of analgesia, which helps a little. He has some shortness of breath on exertion and pain on deep inspiration. He is a nonsmoker and drinks 3 pints of beer a day. He had a transurethral resection of the prostate for prostate cancer 10 years ago and has been discharged from follow-up by the urologist.

- Level of Concern?
- Clinical action?

Spinal Malignancy

- Metastases are cancer lesions that have spread from the primary cancer site to a new and different site in the body.
- Spinal malignancy refers to metastases that have spread specifically into the spine.
- Bone is a common site for metastases, known as metastatic bone disease (MBD), in a number of cancers (breast, prostate, lung, kidney, and thyroid).



Case 6 A.

- A 58-year-old woman with a 42-year history of chronic LBP and history of breast cancer 5 years ago presents with an exacerbation of LBP. No other red flags or signs and symptoms suggestive of a mechanical problem are present on examination.
- History of breast cancer 5 years ago
- Level of Concern?
- Clinical action?

Case 6 B.

- The woman reports that her pain is not responding to usual medication and that she has been prescribed stronger medication, which is helping. She describes her symptoms as different from her usual back pain, which has had some relief from conservative treatment.
- Level of concern?
- Clinical Action

Case 6 C.

- The woman's pain has become progressively worse and she now complains of symptoms waking her at night. She is having difficulty getting back to sleep because the pain is so intense.
- Level of concern?
- Clinical action?

Case 7

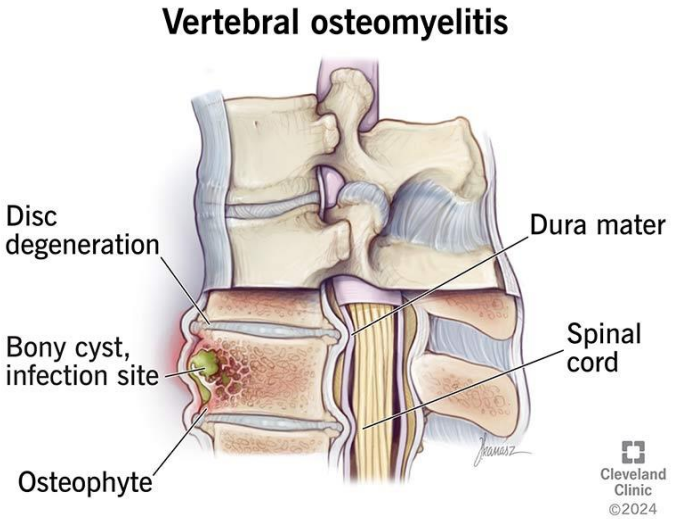
- A 75-year-old man presents with a past history of prostate cancer 2 years ago, and he describes band-like pain and states that his legs feel odd and heavy. He reports occasionally tripping and stumbling. He says he has lost weight but has put it down to a loss of appetite due to the pain.
- Level of concern?
- Clinical Action?

Spinal Infection

Spinal infection is an infectious disease that affects the spinal structures, including the vertebrae, intervertebral discs, and adjacent paraspinal tissues.

In high-income and upper middle–income countries, spinal infection has steadily increased over recent years, possibly due to an aging population and an increase in intravenous drug abuse.

In lower middle–income and lower-income countries, spinal infection has increased due to the dual epidemic of HIV/AIDS and tuberculosis (TB).



Case 8

- A 47-year-old ex-heroin addict presents with recurrent episodes of LBP and a previous history of back pain. He describes symptoms that are intermittent. He is very inactive and usually self-treats the problem with rest. He is an Ex-drug addict.
- Level of concern?
- Clinical action?

Case 9

- A 43-year-old man reports a 3-month history of LBP that is intermittent and mechanical in nature. He was born in Somalia (TB-endemic country) and smokes 20 cigarettes per day. He is neurologically intact and exhibits normal function.
- Level of concern?
- Clinical action?

Case 9 cont.

- The man now feels unwell and has had a fever and chills in the last few days. He reports pain at night and is unable to settle. His pain has now become constant and more intense. Neurological assessment is normal.

- Level of concern?
- Clinical action?

Case 9 cont.

- The man has now developed neurological signs and symptoms, with back and left leg pain to the dorsum of the foot. He scored 3/5 (Oxford strength scale) for left dorsiflexion. He has been up all night with leg pain.

- Level of concern?
- Clinical action?

Thank you